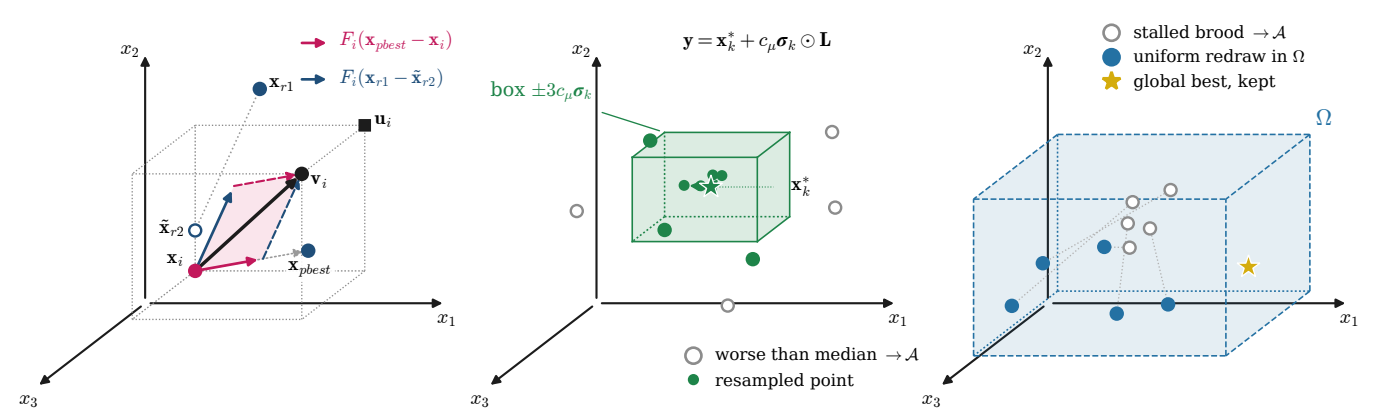
$F_i(\mathbf{x}_{pbest} - \mathbf{x}_i)$
$F_i(\mathbf{x}_{r1} - \tilde{\mathbf{x}}_{r2})$
x_2
$\mathbf{x}_{r1}$
$\mathbf{u}_i$
$\mathbf{v}_i$
$\tilde{\mathbf{x}}_{r2}$
$\mathbf{x}_i$
$\mathbf{x}_{pbest}$
x_1
x_3
a L-SHADE step
$\mathbf{y} = \mathbf{x}_k^* + c_\mu \boldsymbol{\sigma}_k \odot \mathbf{L}$
box $\pm 3 c_\mu \boldsymbol{\sigma}_k$
$\mathbf{x}_k^*$
worse than median $\to \mathcal{A}$
resampled point
b Micro-restart
stalled brood $\to \mathcal{A}$
uniform redraw in Ω
global best, kept
Ω
c Macro-restart